

SMART HOME LIGHT CONTROL SYSTEM

1. Abstract

The Smart Home Light Control System is an IoT-based automation project developed using the ESP32 microcontroller. The system automatically controls a room light using an LDR sensor and a PIR motion sensor. When the environment is dark and human motion is detected, the LED turns ON automatically. A push button allows switching between Automatic and Manual modes, providing flexibility to the user. An OLED display shows the current light condition, motion status, LED status, and operating mode in real time. The project demonstrates the practical application of embedded systems and sensor integration for energy-efficient home automation.

2. Introduction

Home automation has become an essential part of modern living by improving comfort, safety, and energy efficiency. Traditional lighting systems remain ON even when not required, leading to unnecessary power consumption.

This project provides an intelligent lighting solution using ESP32. The LDR sensor measures ambient light intensity, while the PIR sensor detects human movement. Based on these sensor readings, the system automatically controls an LED. A push button allows the user to switch between Auto and Manual modes. The OLED display provides real-time monitoring of system status.

3. Objectives

- Design a smart lighting system using ESP32.
- Detect surrounding light intensity using an LDR.
- Detect human presence using a PIR sensor.
- Automatically control the LED.

- Implement Manual and Automatic operating modes.
 - Display live sensor data on an OLED display.
 - Improve energy efficiency through intelligent lighting.
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4. Project Scope

The project simulates a smart home lighting system capable of controlling lights automatically based on environmental conditions and occupancy. It can be extended with Wi-Fi connectivity for remote monitoring and control using IoT platforms.

5. Components Required

Component	Quantity
ESP32 DevKit V1	1
OLED Display SSD1306	1
LDR Sensor Module	1
PIR Motion Sensor	1
LED	1
Push Button	1
Jumper Wires	As Required
Wokwi Simulator	1

6. Hardware Description

ESP32 DevKit V1

ESP32 is a powerful dual-core microcontroller with built-in Wi-Fi and Bluetooth. It controls all sensors, processes input data, updates the OLED display, and controls the LED.

LDR Sensor

The LDR (Light Dependent Resistor) detects ambient light intensity. It outputs an analog value that is used to determine whether the environment is bright or dark.

PIR Motion Sensor

The PIR sensor detects infrared radiation emitted by humans. When motion is detected, its output becomes HIGH.

OLED Display

The OLED display shows:

- Light Status
- Motion Status
- LED Status
- Operating Mode

Push Button

The push button switches between Auto Mode and Manual Mode.

LED

Represents the room light.

7. Software Requirements

- Arduino IDE

- ESP32 Board Package
- Wokwi Simulator
- Adafruit SSD1306 Library
- Adafruit GFX Library

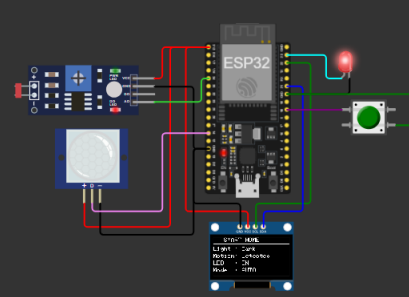
8. Circuit Diagram

```

1  #include <Wire.h>
2  #include <Adafruit_GFX.h>
3  #include <Adafruit_SSD1306.h>
4
5  // OLED display size
6  #define SCREEN_WIDTH 128
7  #define SCREEN_HEIGHT 64
8
9  // Create OLED object
10 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire);
11
12 // Pin connections
13 #define LDR_PIN 34 // LDR Sensor
14 #define PIR_PIN 14 // PIR Motion Sensor
15 #define LED_PIN 23 // LED
16 #define BUTTON_PIN 18 // Push Button
17
18 // Variables
19 bool manualMode = false; // Start in AUTO mode
20 bool ledState = false; // LED initially OFF
21 bool lastButtonState = HIGH; // Store previous button state
22
23 void setup() {
24
25     // Start Serial Monitor
26     Serial.begin(115200);
27
28     // Start I2C communication
29     Wire.begin(21,22);
30
31     // Set pin modes
32     pinMode(PIR_PIN, INPUT);

```

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LED : ON
Mode : AUTO

LDR Value : 448
Motion : Detected
LED : ON
Mode : AUTO

9. Circuit Connections

Component ESP32 GPIO

LDR AO GPIO34

PIR OUT GPIO14

LED GPIO23

Push Button GPIO18

OLED SDA GPIO21

Component ESP32 GPIO

OLED SCL GPIO22

OLED VCC 3.3V

OLED GND GND

10. Working Principle

1. ESP32 reads the analog value from the LDR.
 2. ESP32 reads the digital signal from the PIR sensor.
 3. The push button determines whether the system is in Auto or Manual mode.
 4. In Auto mode:
 - If it is dark and motion is detected, the LED turns ON.
 - Otherwise, the LED remains OFF.
 5. In Manual mode:
 - The LED operation follows the manual control logic implemented in the program.
 6. The OLED continuously displays the current system status.
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11. Algorithm

1. Start the system.
2. Initialize OLED and GPIO pins.
3. Read LDR value.
4. Read PIR sensor status.

5. Read push button status.
 6. Check operating mode.
 7. If Auto Mode:
 - Check light condition.
 - Check motion detection.
 - Control LED.
 8. Update OLED display.
 9. Print data on Serial Monitor.
 10. Repeat continuously.
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13. Program Explanation

The program initializes the OLED display, sensors, and GPIO pins in the setup() function.

Inside the loop() function:

- Reads LDR value
 - Reads PIR sensor
 - Checks button press
 - Changes operating mode
 - Controls LED
 - Updates OLED display
 - Sends data to Serial Monitor
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14. Project Output

The OLED displays:

SMART	HOME
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Light:	Dark
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Motion:	Detected
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LED:	ON
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Mode :	AUTO
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When conditions change, the display updates automatically.

15. Applications

- Smart homes
 - Automatic corridor lighting
 - Office automation
 - Energy-saving systems
 - Smart buildings
 - Home security systems
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16. Advantages

- Low power consumption
- Energy efficient
- Automatic operation
- Easy to install
- Low cost
- Real-time monitoring

- User-friendly interface
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17. Limitations

- Motion detection is limited to the PIR sensor range.
 - LDR performance may vary under different lighting conditions.
 - The project operates locally and does not include cloud connectivity.
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18. Future Enhancements

- Mobile app control using Blynk.
 - Integration with MQTT or cloud platforms.
 - Voice control using Google Assistant or Alexa.
 - Email or notification alerts.
 - Relay-based AC bulb control.
 - Data logging for energy analysis.
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19. Conclusion

The Smart Home Light Control System using ESP32 successfully demonstrates an intelligent lighting solution by combining light sensing and motion detection. The project achieves automatic lighting control, manual operation, and real-time monitoring through an OLED display. It provides practical experience in embedded systems, IoT concepts, sensor interfacing, and energy-efficient automation, making it a suitable project for academic learning and internship work.